Numerical Study of Flow and Heat Transfer in a Centrifugal Pump

Pressure development and thermal response in a rotating annular passage

1. Aim

To investigate pressure generation and temperature changes in a simplified rotating annular passage representing a centrifugal pump. The numerical model is evaluated at different rotational speeds to compare its hydraulic and thermal response.

2. Computational Model and Parameters

The analysis is carried out in a cylindrical coordinate system using a reduced-order model formulated in a rotating polar frame. Radial and tangential velocity components are prescribed, and the model accounts for centrifugal pressure rise, heat exchange with the walls and thermal energy associated with flow losses.

Parameter	Specification
Coordinate system	Cylindrical
Radial grid points	80
Azimuthal grid points	160
Reynolds number	500
Iteration count	1200
Rotational speeds	1000, 2000 and 3000 rpm

3. Boundary Conditions

A prescribed temperature and radial inflow are applied at the pump-eye inlet. The fluid proceeds outward in the radial direction, while periodicity is imposed in the azimuthal direction. Heat transfer at the walls is represented by a distributed source term in the thermal model.

4. Numerical Procedure and Observed Quantities

The calculations are performed for a fixed Reynolds number of 500 while the rotational speed is varied. Pressure contours, estimated pump head, absolute and relative velocity, inlet and outlet temperatures, and thermal-residual histories are used to assess the model response.

5. Results

The values below are reproduced from the supplied report. The additional 5151 rpm case is included because it appears in the simulation plots and the later results discussion.

Speed (rpm)	Re	Inlet temp. (K)	Outlet temp. (K)	Thermal residual	Peak pressure scale
1000	500	300	300.30	3.56 x 10^(-4)	7 x 10^(?)
2000	500	300	300.18	1.81 x 10^(-4)	2.8 x 10^(?)
3000	500	300	300.12	1.22 x 10^(-4)	2.8 x 10^(?)
5151	500	300	300.086	7.1258 x 10^(-5)	1.9581 x 10^6

The residual exponents for the first three cases are readable from the convergence plots. The exponents for the first three peak-pressure entries are unclear in the supplied results table, so they are marked with a question mark rather than guessed.

6. Results and Discussion

The pressure-field plots show a pressure distribution across the annular region, with higher pressure levels toward the outer radius. This pattern is consistent with the centrifugal action produced by rotation. Pressure contours for the different rotational speeds are included in the original simulation pages.

The outlet-temperature histories rise gradually from the prescribed inlet temperature of 300 K. The listed outlet temperatures are 300.30 K at 1000 rpm, 300.18 K at 2000 rpm and 300.12 K at 3000 rpm. The additional 5151 rpm simulation reports an outlet temperature of approximately 300.0862 K.

The thermal-residual curves decrease over the 1200 iterations shown. For the 5151 rpm case, the report gives a final L-infinity residual of 7.1258 x 10^(-5), a pressure rise of 1.9581 x 10^6 Pa and an estimated pump head of about 200.02 m. The reported fluid heat gain is approximately 16.98 W, with a mass flow rate of 0.0471 kg/s.

The source report notes that pump head is expected to scale approximately with the square of rotational speed. A quantitative check of this relationship requires consistent pump-head values for every operating speed; these are not provided for all cases.

7. Conclusion

A simplified centrifugal-pump passage model was used to study pressure development and thermal behaviour at a Reynolds number of 500 and an inlet temperature of 300 K. The simulations compare several rotational speeds and show the corresponding pressure contours, outlet-temperature trends and thermal-residual histories.

For the additional 5151 rpm case, the reported outlet temperature is 300.0862 K, the pressure rise is approximately 1.9581 x 10^6 Pa and the estimated pump head is 200.02 m. The reported heat gain and mass flow rate are 16.98 W and 0.0471 kg/s, respectively.

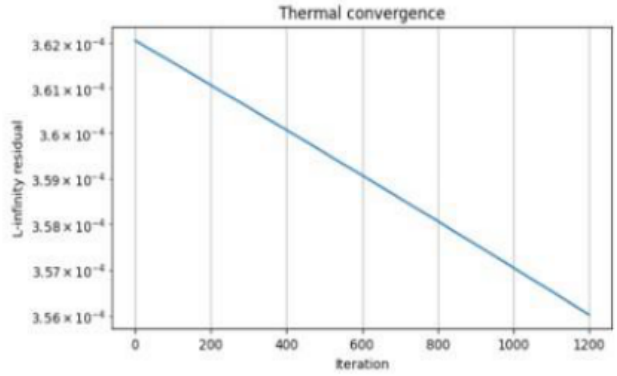
The pressure exponents in the first three rows should be verified against the original solver output. Pump-head values across all speeds would also be needed to assess the expected RPM-squared relationship.

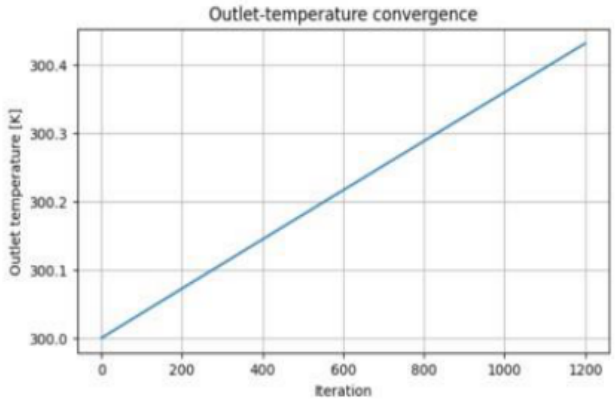
8. Original Simulation Figures

The plots and pressure-field visualizations from the supplied report are retained in the pages that follow.

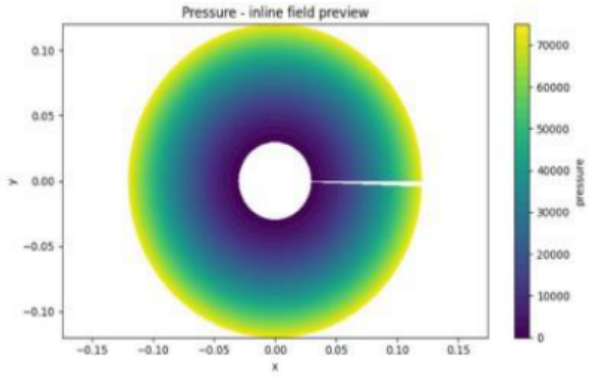
Original simulation output (source page 2)

For pump rotation at 1000 rpm:

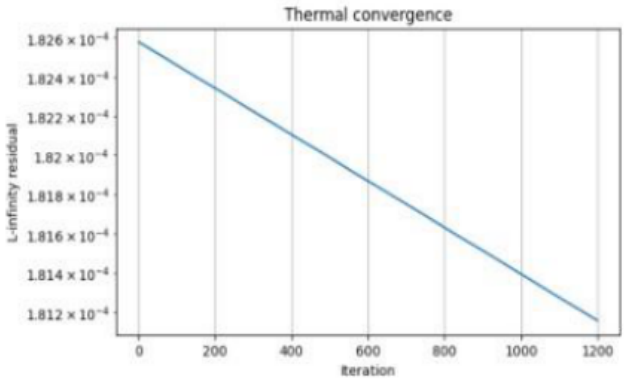




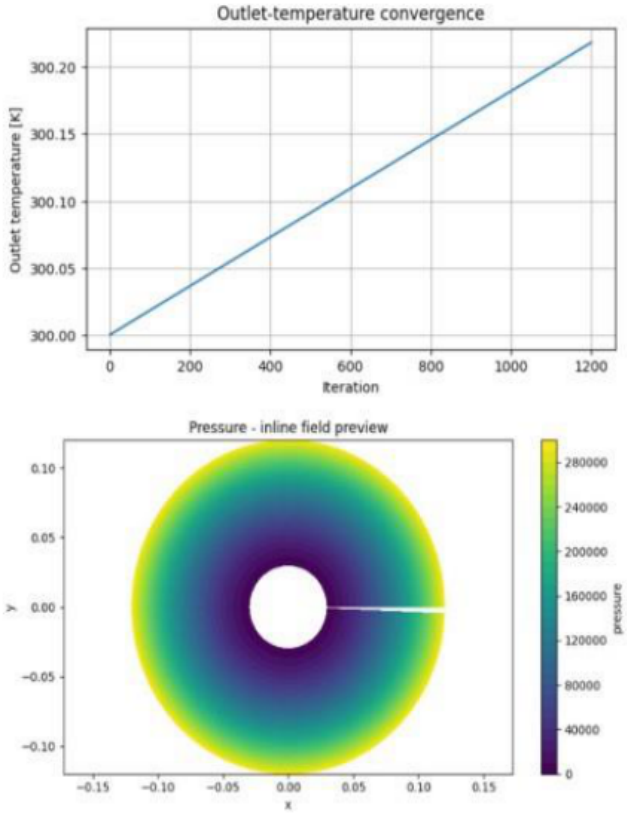
Original simulation output (source page 3)



For pump rotation at 2000 rpm:

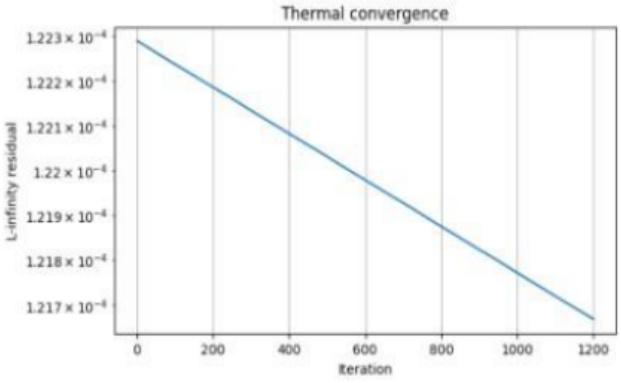


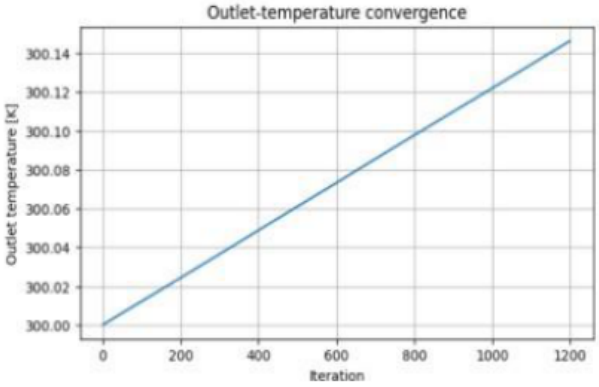
Original simulation output (source page 4)



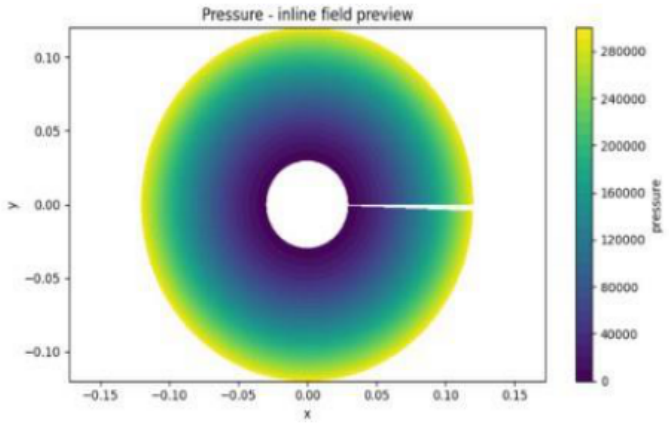
Original simulation output (source page 5)

For pump rotation at 3000 rpm:

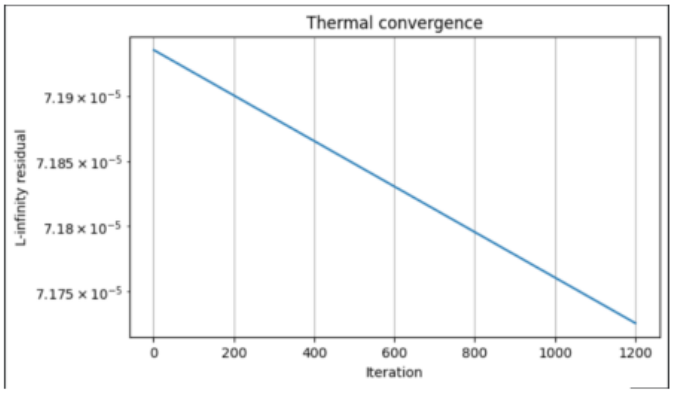


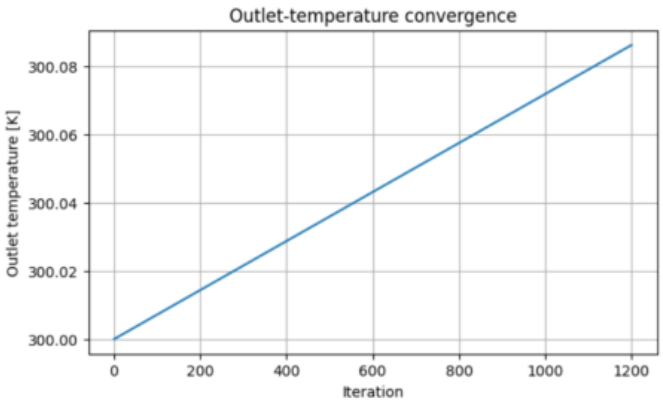


Original simulation output (source page 6)

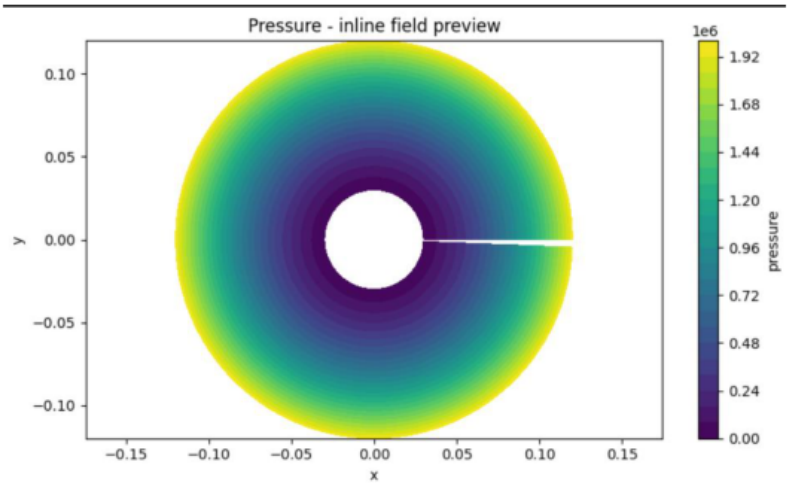


For pump rotation at 5151 rpm:





Original simulation output (source page 7)



Results and Discussion

A simplified two-dimensional annular single-rotating-frame centrifugal pump heat-transfer model was analysed numerically at a fixed Reynolds number of 500. The inlet temperature was maintained at 300 K, while the rotational speed was varied to study its effect on the thermal response and pressure rise of the pump.

At 1000 RPM, the outlet temperature was 300.30 K, with a reported thermal residual of 3.56×10^{-n} and a peak pressure scale of 7×10^{n}. At 2000 RPM, the outlet temperature decreased to 300.18 K, while the thermal residual was 1.81×10^{-n} and the peak pressure scale was 2.8×10^{n}. At 3000 RPM, the outlet temperature was 300.12 K, with a thermal residual of 1.22×10^{n} and a peak pressure scale of 2.8×10^{n}. The exponents in these reported residual and pressure values should be verified from the original solver output.

An additional simulation was performed at 5151 RPM. The solver reported an outlet temperature of 300.0862 K, a pressure rise of 1.9581×10^{6} Pa, a pump head of approximately 200.02 m, and a final L-infinity residual of 7.1258×10^{-5}.

For the 5151 RPM case, the reported fluid heat gain was approximately 16.98 W, and the mass flow rate was approximately 0.0471 kg/s. These results describe the model response at the selected operating condition.

The additional high-speed result provides a further data point for examining the relationship between rotational speed and pump performance. However, the expected near-linear relationship between pump head and RPM² should be evaluated using the pump-head values obtained from all speed cases.

Conclusion

The centrifugal pump heat-transfer model was investigated at a fixed Reynolds number of 500 and an inlet temperature of 300 K for different rotational speeds. The reported outlet temperatures for 1000, 2000, and 3000 RPM were 300.30 K, 300.18 K, and 300.12 K, respectively. For the additional case at 5151 RPM, the solver predicted an outlet temperature of 300.0862 K.

At 5151 RPM, the numerical model reported a pressure rise of approximately 1.9581 MPa, a pump head of approximately 200.02 m, and a final L-infinity residual of 7.1258×10^{-5}. The reported fluid heat gain was approximately 16.98 W.

Original simulation output (source page 8)

The results demonstrate how the simplified numerical model can be used to investigate the thermal and pressure-related performance of a centrifugal pump at different rotational speeds. A quantitative assessment of pump-head scaling with RPM² requires consistent pump-head results for all operating speeds. The reported values should also be checked against the original solver output before final submission, particularly the thermal residuals and peak-pressure values for the first three cases.